VEGO-AI | PHD RESEARCH PROGRAMME

Chapter 2 - Literature Review: Human-Agent Collaboration and Governed Judgment

Evidence-bounded draft for supervisor review - literature only

Evidence boundary: 525 occurrences -> 116 works; human screening 0/116; EXP-005 0/24; medical readiness 0/6.

Status: Ready for Ali review; not delivered, accepted, or closed.

C2-01 Review scope and evidence method

This chapter is a bounded literature tranche prepared in response to the closing instruction from the 12 August supervision call: advance the literature first and defer further methodology development until the literature has progressed. The pre-existing Section 4 methodology draft is therefore frozen and is neither extended nor treated as approved in this package.

The evidence base has four levels. First, the ACL 2026 survey is used with page-level anchors for survey-level statements. Second, its associated taxonomy repository is pinned to commit 7b3ba9deefe99172748582f6025d995ccc2a6f86 and parsed deterministically. Third, twelve works are listed only as conservative candidates for human title, abstract, and full-text review. Fourth, the separate local corpus, native workbook, and one live query test are recorded as process evidence rather than as a completed review.

The pinned repository contains 525 in-scope occurrences that resolve to 116 distinct works. Deterministic title and taxonomy mapping is complete, but human screening is 0/116. The preliminary machine mapping classifies 11 works as core candidates, 40 as relevant, 47 as contextual, and 18 as peripheral. These labels prioritize review; they do not establish inclusion, quality, identity, or findings.

QL-01 through QL-05 remain protocol-ready and unexecuted. An unverified operator observation from the Google Scholar Foundations test reported about 2,500 displayed results and no CAPTCHA on 15 August. No export or screening occurred, and neither the count nor the query wording has been independently confirmed. This observation does not satisfy readiness and is not counted as formal-query execution.

Evidence layer	Current coverage	Permitted use
ACL survey	Primary source with six page-level anchors	Survey-level taxonomy statements
Pinned repository	525 occurrences; 116 works	Title and taxonomy metadata only
Candidate set	12 conservative starting points	Prioritization for human screening
Formal searches	QL-01-QL-05 unexecuted	Protocol status only

Source anchors: C2-ACL-01; C2-ACL-02

C2-02 Problem setting: operationalized guidelines and variability

The proposal's working problem concerns assessment of operationalized guidelines and the variability that appears when people, teams, or domains instantiate those guidelines differently. This is a project problem statement, not a literature finding from the bounded ACL corpus. Domain-modeling and guideline-operationalization sources still require targeted human screening before the chapter can establish the breadth, prevalence, or consequences of that variability.

Within this working problem, human judgment can enter at three analytically distinct points: deciding when intervention is warranted, determining how a judgment and its basis are represented and governed, and evaluating whether later use or transfer remains valid. These are organizing lenses for the review, not approved research-question wording. The unresolved choices between exploration and identification and between human and expert terminology remain visible decisions rather than silently fixed vocabulary.

The software and modeling artifacts already produced by VEGO-AI provide preliminary mechanism evidence only. They do not establish improved accuracy, generalization, expert-effort reduction, clinical value, or doctoral novelty. EXP-005 remains at 0/24 independent safe labels and medical readiness remains at 0/6 gates.

C2-03 Human-agent collaboration design space

Zou et al. describe human-agent systems through environment and profiling, feedback, interaction, orchestration, and communication. Their model assigns humans roles involving information or clarification, feedback or correction, and control or action. This survey-level architecture provides a useful vocabulary for comparing how human participation is configured without assuming that any one configuration is effective [C2-ACL-01; C2-ACL-02].

The associated repository makes four operational branches explicit. Human Feedback captures what information or control is supplied and when; Interaction distinguishes collaboration, competition, and coopetition and refines collaboration into delegation, supervision, cooperation, and coordination; Orchestration separates task strategy from temporal synchronization; and Communication separates structure from mode [C2-ACL-03; C2-ACL-04; C2-ACL-05].

The survey also proposes a five-level Human Agency Scale from full automation to human-driven work [C2-ACL-06]. For this research, the scale is a candidate analytical aid rather than an adopted measure. Its construct validity, fit with domain experts, and relationship to workload or authority must be appraised before use.

Taxonomy branch	Encoded dimensions	Use in this review
Human Feedback	Type, subtype, granularity, phase	Characterize what humans provide and when
Interaction	Type and collaboration variant	Distinguish delegation, supervision, cooperation, coordination
Orchestration	Strategy and synchronization	Separate task ordering from temporal control
Communication	Structure and mode	Separate who communicates from how

Source anchors: C2-ACL-01; C2-ACL-02; C2-ACL-03; C2-ACL-04; C2-ACL-05; C2-ACL-06

C2-04 Selective intervention and bounded oversight

The survey's feedback taxonomy distinguishes type, granularity, and phase, which supports a structured account of when human input enters an agentic workflow [C2-ACL-03]. It does not by itself determine which uncertainty signal should trigger intervention, how importance should be defined, or how review burden should be measured. Those are empirical and design questions that require primary-study review and an explicit evaluation protocol.

Four repository works are high-priority candidates for this section because their titles or taxonomy positions concern clarification seeking, proactive intervention, controlled autonomy, or oversight work (C2-KS-02 through C2-KS-05). At this stage the chapter attributes no method, result, or conclusion to them. Their independent identity, abstracts, study design, and full text remain to be reviewed by a human.

A defensible synthesis must separate system routing counts from measured human effort. Queue volume, trigger frequency, and modeled review rate can characterize mechanism behavior; they cannot establish reduced burden without real reviewers, time-on-task, and an agreed comparator.

Source anchors: C2-ACL-03; C2-KS-02; C2-KS-03; C2-KS-04; C2-KS-05

C2-05 Feedback, judgment representation, governance, and reuse

Feedback categories describe what a human supplies to a system, but governed reuse requires additional properties: case grounding, provenance, accountable authority, validation and adjudication, conflict handling, scope, expiry, revocation, and a traceable rationale for later retrieval. These properties are not explicit dimensions in the pinned repository taxonomy schema. This is a schema-coverage observation, not a claim that no paper in the corpus discusses them.

The titles and taxonomy positions of MINT, preference learning from user edits, and collaborative memory with access control make C2-KS-07 through C2-KS-09 reasonable candidates for this section. No finding is attributed before identity verification and full-text screening.

The proposed governed-judgment lifecycle therefore remains a contribution hypothesis. The literature review must test whether prior work already represents these properties, how authority and disagreement are handled, and whether reuse is evaluated across genuinely separated contexts. Until that screening is complete, statements such as 'no framework exists' are not permitted.

Source anchors: C2-ACL-03; C2-KS-07; C2-KS-08; C2-KS-09

C2-06 Evaluation, transfer, and context boundaries

The bounded corpus includes candidate benchmarks and evaluation frameworks for human-agent participation and domain-specific collaboration (C2-KS-06 and C2-KS-10 through C2-KS-12). Their presence shows that evaluation and human participation are represented in the corpus; it does not establish that any of them evaluates reuse of governed judgment across operationalized-guideline contexts.

Transfer must be decomposed before it can be claimed. System behavior may transfer while a particular judgment does not; a judgment may remain valid within a diagram family but not across domains; and agreement may reflect shared ambiguity rather than correct generalization. The literature review must therefore record unit of transfer, context boundaries, comparator, reference standard, subgroup effects, and leakage controls for every included study.

Plan A and Plan B wording remains open. Healthcare is a conditional transfer context, not an established evidence source, and no patient-row work is authorized. With medical readiness at 0/6, this chapter uses no clinical-performance or partner-access claim. Under the protected fallback, every research question must remain answerable using software or modeling evidence and an authorized non-medical replication context.

Source anchors: C2-KS-06; C2-KS-10; C2-KS-11; C2-KS-12

C2-07 Synthesis and candidate gaps

The current evidence supports a bounded synthesis: the survey provides a mature vocabulary for feedback, interaction, orchestration, communication, and levels of human agency; the pinned repository provides a reproducible candidate corpus; and the taxonomy schema does not explicitly encode several governance and transfer dimensions central to the working problem.

Four candidate gaps remain hypotheses pending human screening: how intervention timing is linked to independently defined importance; how judgments are grounded, validated, reconciled, scoped, and revoked; when a judgment artifact can be reused across contexts; and how expert burden is measured rather than projected. The absence of these fields from the taxonomy schema is evidence about schema coverage only. It is not evidence that the underlying literature lacks relevant methods.

The contribution claim must therefore remain conditional. If full-text review finds existing tested approaches covering these dimensions, the proposal should narrow, absorb, or supersede the corresponding contribution. If the dimensions remain weakly connected after review, the proposal may justify a design-science contribution, but only with a source-backed gap matrix and falsifiable evaluation plan.

Candidate gap	Current evidence state	Required closure evidence
Intervention timing and importance	Pending full-text review	Primary studies plus an independent importance definition
Provenance, authority, conflict and revocation	Not explicit in taxonomy schema	Full-text coding across included works
Safe judgment reuse across contexts	Not explicit in taxonomy schema	Transfer-unit and context-boundary evidence
Measured expert burden	Not explicit in taxonomy schema	Human time-on-task and workload measures

Source anchors: C2-ACL-01; C2-ACL-03; C2-ACL-04; C2-ACL-05

C2-08 Next literature tranche and source register

The recommended next task is a controlled human screening tranche over the twelve conservative candidates. For each work, verify bibliographic identity, review title and abstract, record include or exclude with a reason, and review full text only when included. Authors' conclusions and Ali's synthesis must remain separate fields in the native workbook.

The native workbook remains the canonical literature record and is referenced here only by the opaque alias VEGO-AI-PHD-LITERATURE-WORKBOOK. Its live link is withheld from this tracked review package and may be injected only from an ignored private binding after Ali approves the exact release package. The 15 August snapshot contains six paper rows; the 116 staged corpus rows have not been written. Any append requires Ali approval, a fresh header and row-count check, and corrected restricted sharing before recipient access testing.

The broader QL-01 through QL-05 searches remain deferred. The next supervisor decision is whether this bounded corpus plus the twelve-work screening tranche is sufficient for the proposal-stage Chapter 2, or whether a specific database query must be advanced before methodology resumes.

Source anchors: C2-ACL-01; C2-ACL-02; C2-ACL-03; C2-ACL-04; C2-ACL-05; C2-ACL-06

Sources and evidence anchors

The ACL anchors below were reviewed at page level. The twelve additional works remain human-screening candidates only.

C2-ACL-01 | p. 36335, Abstract | The survey frames LLM-based human-agent systems as incorporating human information, feedback, or control and organizes the field around environment and profiling, feedback, interaction, orchestration, and communication.

Boundary: Survey-author framing; not proof of improved performance, reliability, or safety. https://aclanthology.org/2026.findings-acl.1811/

C2-ACL-02 | p. 36336, Figure 1 and Section 2 | The survey model identifies five system components and human roles involving information or clarification, feedback or correction, and control or action.

Boundary: Taxonomy backbone only; listed benefits are not VEGO-AI results. https://aclanthology.org/2026.findings-acl.1811/

C2-ACL-03 | pp. 36337-36338, Section 3.2 and Table 1 | Human feedback is categorized by type, granularity, and phase, including evaluative, corrective, guidance, and implicit forms.

Boundary: Category definitions are source-reported; individual corpus findings remain unscreened. https://aclanthology.org/2026.findings-acl.1811/

C2-ACL-04 | pp. 36339-36340, Section 3.3, Figure 2 and Table 2 | Interaction is separated into collaboration, competition, and coopetition; collaboration is refined into delegation, supervision, cooperation, and coordination.

Boundary: One survey taxonomy, not the only possible classification. https://aclanthology.org/2026.findings-acl.1811/

C2-ACL-05 | pp. 36340-36341, Sections 3.4-3.5 and Table 2 | Orchestration distinguishes task strategy from temporal synchronization, while communication distinguishes structure from mode.

Boundary: Implementation effectiveness cannot be inferred without primary-study review. https://aclanthology.org/2026.findings-acl.1811/

C2-ACL-06 | pp. 36341-36342, Section 3.6 and Table 3 | The survey adds a five-level Human Agency Scale spanning full automation through human-driven work.

Boundary: Scale selection and suitability require critical appraisal. https://aclanthology.org/2026.findings-acl.1811/

Conservative screening candidates

C2-KS-01 | LLM-Based Human-Agent Collaboration and Interaction Systems: A Survey | Taxonomy backbone | https://arxiv.org/abs/2505.00753

C2-KS-02 | Ask or Assume? Uncertainty-Aware Clarification-Seeking in Coding Agents | Clarification candidate | https://arxiv.org/abs/2603.26233

C2-KS-03 | Ask-before-Plan: Proactive Language Agents for Real-World Planning | Proactive-intervention candidate | https://aclanthology.org/2024.findings-emnlp.636/

C2-KS-04 | A Decoupled Human-in-the-Loop System for Controlled Autonomy in Agentic Workflows | Bounded-autonomy candidate | https://arxiv.org/abs/2604.23049

C2-KS-05 | Human Oversight of Agentic Systems in Practice | Oversight-work candidate | https://arxiv.org/abs/2606.05391

C2-KS-06 | Collaborative Gym: A Framework for Enabling and Evaluating Human-Agent Collaboration | Evaluation candidate | https://arxiv.org/abs/2412.15701

C2-KS-07 | MINT: Evaluating LLMs in Multi-turn Interaction with Tools and Language Feedback | Feedback-environment candidate | https://arxiv.org/abs/2309.10691

C2-KS-08 | Aligning LLM Agents by Learning Latent Preference from User Edits | Judgment or preference candidate | https://openreview.net/forum?id=DlYNGpCuwa

C2-KS-09 | Collaborative Memory: Multi-User Memory Sharing in LLM Agents with Dynamic Access Control | Memory and access-control candidate | https://arxiv.org/abs/2505.18279v1

C2-KS-10 | HAS-Bench: Evaluating LLM-Based Human-Agent Systems under Configurable Human Participation | Participation-evaluation candidate | https://arxiv.org/abs/2607.04329

C2-KS-11 | SPHERE: An Evaluation Card for Human-AI Systems | Evaluation-card candidate | https://arxiv.org/abs/2504.07971

C2-KS-12 | AgentDS: Benchmarking the Future of Human-AI Collaboration in Domain-Specific Data Science | Domain-transfer candidate | https://arxiv.org/abs/2603.19005

Native literature workbook: VEGO-AI PhD Literature Workbook v0.1 [VEGO-AI-PHD-LITERATURE-WORKBOOK] | Live link withheld pending Ali-approved release